MAT627 Chapter 2.6 Programming Assignment

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1 Problem Setup

We solve

$$T\mathbf{x} = \mathbf{b},$$

where $T \in \mathbb{R}^{n\times n}$ is tridiagonal with

$$t_{ij} = \begin{cases} 1, & |i-j| = 1, \\ i+1, & i = j, \\ 0, & \text{otherwise}, \end{cases} \qquad b_i = 1.$$

The implementation stores only the lower diagonal ℓ, diagonal d, upper diagonal u, right-hand side $\mathbf{b}$, and solution $\mathbf{x}$, then applies Algorithm 2.6 (Thomas elimination and back substitution). After solving, it computes the residual

$$\mathbf{r} = \mathbf{b} - T\mathbf{x}$$

using tridiagonal matrix-vector multiplication and reports

$$e_n = \max_{1\le i\le n} |r_i|.$$

Table 1: Maximum residual component for the tridiagonal solves.

n	$e_n = \max_i \lvert r_i \rvert$
3	0.000000×10^{0}
10	0.000000×10^{0}
25	2.220446×10^{-16}
100	2.220446×10^{-16}

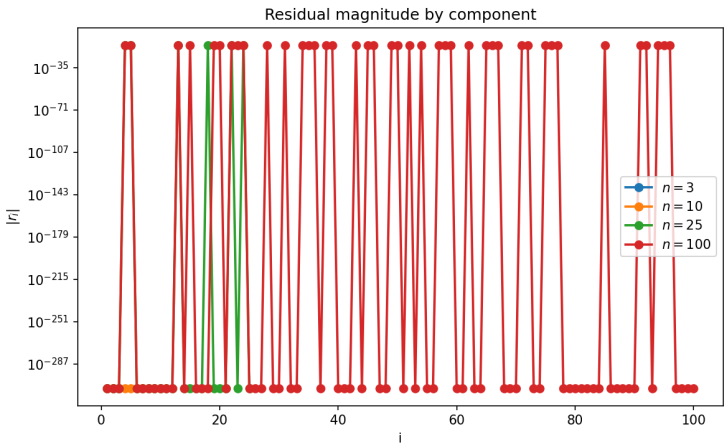

Figure 1: Residual component magnitudes for the four test systems.

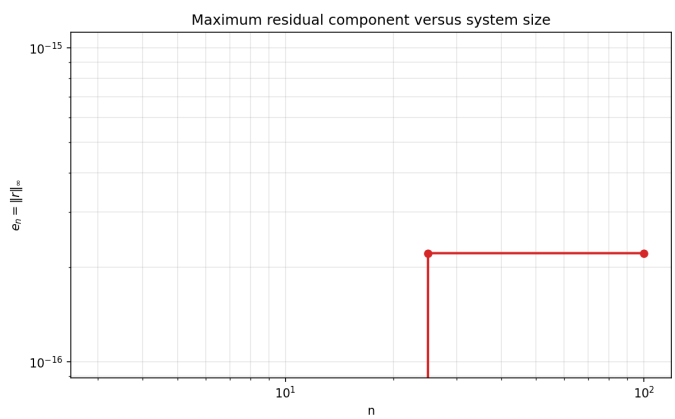

Figure 2: Maximum residual magnitude as a function of system size.